

Should We Keep the Policy? Forward-Looking Estimands and Identification for Panel Data

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Abstract

Most causal inference rests on an implicit assumption we call the Fundamental Promise of Causal Inference: that effects estimated on data from the past can inform decisions in the present and future. Yet the recent panel-data literature emphasizes *dynamic* effects that change over time, and methods that allow such dynamics estimate backward-looking aggregates—averages over already-observed periods—that need not answer the forward-looking question a policymaker actually faces: what will the effect be if we keep, repeal, or adopt this policy going forward? We formalize this gap with a dual contribution: (1) we introduce policy-relevant *future causal effects*—the Future ATT, Future ATU, and Future ATE, mapping respectively to repeal, adoption, and universal-adoption decisions; and (2) we give three identification strategies that extrapolate from the data we can analyze (the past) to the estimands we care about (the future), differing in what they assume is invariant. Path 1 assumes time homogeneity; Path 2 specifies parametric temporal dynamics; Path 3 holds the covariate-conditional effect fixed and transports it across a shifting covariate distribution, remaining valid under regime change. Each path plugs an existing estimator into a forward map and propagates its influence function for valid inference; we also adapt time-series cross-validation to model selection for Path 2. An application to Stand-Your-Ground laws illustrates both the promise and limits of extrapolation. Software: R package `extrapolateATT`.

1 Introduction

There is a fundamental and unacknowledged tension at the heart of policy analysis[Egami and Hartman, 2023, Heckman and Vytlačil, 2001] – and causal inference with panel data more broadly. This tension arises because of the possibility of dynamic causal effects—effects that change over time. If policy effects are truly unrestricted in their dynamics, historical data cannot inform any current policy decisions: yet informing current decisions is presumably the reason we estimate causal effects in the first place.

Most causal inference rests on an implicit assumption of time homogeneity. We call this assumption—our label—the Fundamental Promise of Causal Inference: causal effects estimated (by necessity) on data collected in the past can be used

to inform decision-making in the present [Egami and Hartman, 2023, Khosrowi, 2019, Cartwright, 2007]. If two identical clinical trials – same target population and inclusion criteria, same treatment, same outcome – were expected to not only give different results but in fact target different causal effects simply because of the passage of time, that would be a major crisis for causal inference. A durable understanding of causal effects would be elusive.

Public policies – and most causal inference studies on panel data – are widely understood to violate this assumption and are assumed to have effects that inherently change over time [Lucas Jr, 1976, Ericsson et al., 1998, Estrella and Fuhrer, 1999]. Policies are introduced into dynamic contexts, may be administered or enforced differently over time, and may themselves cause changes in public behavior and attitude. Technological advancement, public opinion, economic conditions, and new research may all modify the causal effects of policies.

This basic assumption of dynamic effects has driven a great deal of recent advances in methods for causal inference with panel data [Wang et al., 2024]. Recent surveys clarify how heterogeneity and dynamics enter identification and estimation in this setting [Arkhangelsky and Imbens, 2024, De Chaisemartin and d’Haultfoeuille, 2023]. It is now standard to assume that causal effects vary both with time of treatment and time since implementation. These effects (often called “group-time” effects) may be of interest in themselves but more typically are aggregated into coarser effects like an “overall” average treatment effect on the treated (ATT). Assuming such dynamic effects solves a problem with classical methods like two-way fixed effects (TWFE): TWFE may incur bias in the presence of time-heterogeneous effects because they implicitly include comparisons between units treated at different times in their estimate of the ATT [Sun and Abraham, 2021].

While methods assuming dynamic effects have been developed for good reasons, they strike at the heart of the Fundamental Promise of Causal Inference. In the presence of dynamic effects, it is not clear how one can connect causal effects estimated on historical data to any decision one would make now or in the future [Egami and Hartman, 2023, Devaux and Egami, 2022, Khosrowi, 2019]. Unfortunately, policy evaluation is fundamentally future-oriented. A policymaker deciding whether to maintain, repeal, or adopt a policy only cares about effects going forward: What will happen if we keep this policy in place? What if we implement it in new jurisdictions?

The question of when and how historical evidence can inform current or future decisions is central to the literature on generalizing and transporting causal effects [Cole and Stuart, 2010, Tipton, 2013, Manski, 2013]. If one estimates the effect of a public policy to be beneficial (e.g., improves life expectancy) among states that implemented it in historical data, does that mean it would be wise to maintain the policy where it is already in place, or to expand it to jurisdictions where it has not? If effects are unrestrictedly dynamic, it’s impossible to say.

We seek to resolve this tension between dynamic effects and the Fundamental Promise by encouraging researchers to either take the possibility of dynamic effects *less* seriously or *more* seriously—and, when taking them more seriously, to choose what they believe is invariant. Taking it less seriously (Path 1) involves assuming some level of time homogeneity in causal effects. This would allow causal effects that would inform current and future decisions to be identified in historical data. We argue that an informal version of this is currently standard practice, where average treatment effects estimated on historical data

are treated as if they are stable enough over time to inform current debates. The notion of policy-relevant treatment effects formalizes which estimands answer counterfactual policy questions [Heckman and Vytlačil, 2001, Sasaki and Ura, 2023]. On the other hand, taking dynamic effects more seriously would require formally modeling how effects change over time. Under Path 2, researchers specify temporal dynamics (e.g., trends in event time) and extrapolate [Khosrowi, 2019, Maeba, 2024]. Under Path 3, researchers model structural heterogeneity via covariates and integrate over the future covariate distribution, connecting to structural methods that enable extrapolation across regimes [Lucas Jr, 1976, Heckman and Vytlačil, 2005, Pearl and Bareinboim, 2022].

We introduce new causal estimands that formalize the notion that causal inference is inherently future-oriented, which we call “future causal effects.” We locate the estimands outside the time range of the current study and present three identification paths that differ in what they assume is invariant: Path 1 assumes temporal stability of effects; Path 2 assumes a functional form for temporal dynamics; Path 3 assumes structural covariates remain predictive under regime change. Across fields, three broad strategies emerge for making causal claims about the future: structural or parameter invariance (e.g., policy-invariant deep parameters); statistical transport or reweighting of effects across populations; and explicit modeling of heterogeneity or dynamics. Our three paths connect to these traditions: Path 1 invokes parameter invariance (time homogeneity as structural stability); Path 2 relies on explicit modeling of temporal dynamics; Path 3 combines structural parameter stability with reweighting over the future covariate distribution, most directly engaging the Lucas critique.

Related Work

Panel Data and Difference-in-Differences Our setting builds on recent advances in causal inference for panel data. Comprehensive surveys by Arkhangelsky and Imbens [2024] and De Chaisemartin and d’Haultfoeuille [2023] clarify how heterogeneity and dynamics enter identification and estimation. The group-time and staggered adoption literature [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Roth and Sant’Anna, 2023] addresses inference for backward-looking effects when treatment timing varies. Work on testing parallel trends [Freyaldenhoven et al., 2019, Roth, 2022, Rambachan and Roth, 2023, Dette and Schumann, 2024] connects to our testable implications of time homogeneity. These lines focus on effects in the observed window; we target estimands relevant for current and future policy decisions.

External Validity and Transportability Our forward-looking estimands extrapolate across time, which is closely connected to the literature on extrapolating across populations. Egami and Hartman [2023] provide a framework for external validity; Pearl and Bareinboim [2022] and Bareinboim and Pearl [2012] give identification results for transporting effects across settings, and Cole and Stuart [2010], Tipton [2013] for generalizing from trials to target populations. Khosrowi [2019, 2023] examine the assumptions underlying extrapolation, and Manski [2013] advocates bounded identification when those assumptions are weak. Our Path 3 makes this connection concrete: it transports a covariate-conditional effect across the shift in covariate distribution from the observed period to the future, treating time as the dimension along which populations differ.

Lucas Critique and Structural Approaches Path 3’s logic is that of the Lucas Critique [Lucas Jr, 1976, Ericsson et al., 1998, Estrella and Fuhrer, 1999]: reduced-form relationships estimated under one regime need not survive a change of regime, but structural parameters that are invariant to the intervention can. Structural and equation-based approaches [Heckman and Vytlačil, 2005, Pearl and Bareinboim, 2022] show how holding such parameters fixed enables extrapolation across regimes; we operationalize this by holding the covariate-conditional effect fixed while the covariate distribution shifts.

Policy-Relevant Estimands and Forecasting Heckman and Vytlačil [2001] define policy-relevant treatment effects for counterfactual policy changes, and Sasaki and Ura [2023] provide estimation and inference for such parameters. Closest to our aims, Forastiere et al. [2025] formalize forecasting the causal effect of *re-implementing* an intervention, and Gische et al. [2021] distinguish forecasting the effect of an intervention from predicting future outcomes; Deb et al. [2026] forecast unit-level counterfactual outcomes under a factor model. We differ in targeting population-level future causal effects (FATT/FATU/FATE) at a generic horizon, under transparent invariance assumptions, with influence-function-based inference.

2 Setting and Notation

2.1 Panel Data Framework

We observe n units over p time periods. For unit $i = 1, \dots, n$ and time $t = 1, \dots, p$, let $A_{it} \in \{0, 1\}$ denote a policy indicator, Y_{it} a scalar outcome of interest, and X_i time-invariant pre-treatment covariates. Define $Y_{it}(a)$ as the potential outcome for unit i at time t if the policy were set to $A_{it} = a$, possibly contrary to fact. The following assumption links observed outcomes to potential outcomes.

Assumption 1 (Consistency). $Y_{it}(a) = Y_{it}$ whenever $A_{it} = a$.

Consistency precludes interference and multiple versions of treatment: the potential outcome under a is uniquely defined and equals the observed outcome when unit i actually receives treatment a at time t .

Let $\mathcal{O} = \{X_i, Y_i(0), Y_i(1), A_i\}_{i=1, \dots, n} \sim \mathbb{P}$, where $Y_i(a) = (Y_{i1}(a), \dots, Y_{ip}(a))^\top$ collects potential outcomes through period p and $A_i = (A_{i1}, \dots, A_{ip})^\top$ collects treatment indicators. Treatment indicators may change over time according to *staggered adoption*: units with $A_{it} = 0$ may have $A_{is} = 1$ for some $s > t$, but we assume that $A_{it} = 1$ implies $A_{is} = 1$ for all $s > t$. Let $G_i = \min\{t : A_{it} = 1\}$ be the time of adoption, which is often called the *group* or *cohort*. We write $\mathbb{P}_t$ for the marginal distribution of $(X_i, Y_{it}(0), Y_{it}(1), A_{it})$ at time t . Crucially, we also define $\mathbb{P}_{p+k}$ for integer $k \geq 1$ as the distribution of $(X_i, Y_{i,p+k}(0), Y_{i,p+k}(1), A_{i,p+k})$ in the same superpopulation k periods ahead—formally, the marginal at $t = p + k$ of the joint process governing the data through $p + k$. These future marginals are not directly observed but are the target of the estimands in Section 3.

2.2 Staggered Adoption and Group-Time ATTs

In staggered adoption settings, the standard summary estimand is the average treatment effect on the treated (ATT),

$$\theta = \mathbb{E}_{\mathbb{P}}\{Y_{it}(1) - Y_{it}(0) \mid A_{it} = 1\}, \quad (1)$$

which averages over all treated units and time periods in the observed window. In the presence of dynamic effects—where impacts vary across calendar time and time since implementation—it is important to decompose the ATT into finer components based on time of adoption and time of measurement: the group-time ATT is

$$\theta_{gt} = \mathbb{E}_{\mathbb{P}_t}\{Y_{it}(1) - Y_{it}(0) \mid A_{it} = 1, G_i = g\}, \quad (2)$$

the average effect at calendar time t for units in cohort g [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021]. These sub-ATTs isolate variation in effects across cohorts and over time; they are typically aggregated into θ or a weighted summary akin to θ . Modeling effects at this level of granularity also avoids the well-known bias of TWFE estimators, which implicitly compare units treated at different times and can yield sign-reversed estimates under treatment effect heterogeneity [Sun and Abraham, 2021].

All of these estimands— θ and θ_{gt} —are defined over the observed window $t \leq p$. They describe what has already happened. Our forward-looking estimands (Section 3) instead target effects at future times $t > p$, which requires extrapolating from the observed $\{\theta_{gt}\}_{g,t \leq p}$ to periods not yet in the data.

2.3 First-Stage Identification

We do not restrict the first-stage strategy for identifying θ_{gt} ; this paper does not propose new first-stage designs. We give two example designs and their attendant assumptions to show how these effects are typically identified in practice, but certainly many more options occur.

Design 1: Difference-in-differences.

Assumption 2 (Overlap (DiD)). *For each cohort g and time $t \leq p$, $\mathbb{P}(G_i > t \mid G_i \geq g) > 0$.*

Assumption 3 (Parallel trends). *For all g, t , there exists at least one $s < g$ such that*

$$\mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid G_i = g] = \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid G_i > t].$$

Under Assumptions 1–3, θ_{gt} is identified for all $g \leq t \leq p$ [Callaway and Sant’Anna, 2021].

Design 2: Unconfoundedness. For this design, let $\mathcal{C}_g \subset \{X_i, Y_{i1}, \dots, Y_{i,g-1}\}$ denote a set of confounders for cohort g ; this may include the time-invariant covariates, pre-treatment outcomes, or both.

Assumption 4 (Overlap (unconfoundedness)). *For each cohort g , $0 < \mathbb{P}(G_i = g \mid \mathcal{C}_g, G_i \geq g) < 1$ almost surely.*

Assumption 5 (No unmeasured confounding). *For each cohort g , $\{Y_{it}(0)\}_{t=1}^p \perp\!\!\!\perp G_i \mid \mathcal{C}_g, G_i \geq g$.*

In synthetic control studies $\mathcal{C}_g$ is typically taken to be the pre-treatment outcomes $Y_{i1}, \dots, Y_{i,g-1}$; in observational studies it is more commonly taken to be X_i . Under Assumptions 1, 4, and 5, θ_{gt} is identified via inverse probability weighting or outcome regression [Callaway and Sant’Anna, 2021].

This paper’s contribution begins where first-stage identification ends. Given identified $\{\theta_{gt}\}_{g,t \leq p}$, when and how do these backward-looking estimands identify the forward-looking FATT and FATE? The full identification argument thus has two steps: (a) under the researcher’s first-stage design, θ_{gt} is identified from observed data; (b) under the extrapolation assumptions of Section 4, FATT and FATE are identified from θ_{gt} . The three paths in Section 4 differ only in what they assume for step (b).

2.4 Efficient Influence Functions for First-Stage Estimators

To conduct valid inference for the forward-looking estimands, we need to propagate estimation uncertainty from the first-stage estimates $\hat{\theta}_{gt}$ through to the final estimand. We do this via the efficient influence function (EIF) framework.

We work at a level of abstraction that does not commit to a particular first-stage design. We assume only that each first-stage estimator $\hat{\theta}_{gt}$ is *asymptotically linear*: there exists a mean-zero function $\varphi_{gt}(O_i)$ satisfying $\mathbb{E}[\varphi_{gt}^2] < \infty$ such that

$$\sqrt{n}(\hat{\theta}_{gt} - \theta_{gt}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi_{gt}(O_i) + o_{\mathbb{P}}(1). \quad (3)$$

Here O_i denotes all observed data for unit i . The function φ_{gt} is the *influence function* of $\hat{\theta}_{gt}$; when $\hat{\theta}_{gt}$ is a semiparametric efficient estimator, φ_{gt} is the efficient influence function for θ_{gt} under the first-stage design. Asymptotically linear estimators exist for both DiD designs [Callaway and Sant’Anna, 2021] and unconfoundedness designs; the specific form of φ_{gt} depends on which design is used, but the extrapolation theory in Sections 4–5 requires only (3).

Given an asymptotically linear first-stage estimator of the relevant object, standard semiparametric theory yields an asymptotically linear estimator of the forward-looking FATT for each identification path. For Paths 1 and 2 the relevant object is the group-time ATTs θ_{gt} and the FATT is a smooth, finite-dimensional functional of (θ_{gt}, ω_g) , so its influence function follows by the chain rule and functional delta method (Section 5). For Path 3 the relevant object is the covariate-conditional effect $\tau(\cdot)$, which is estimated nonparametrically and need not be $\sqrt{n}$ -consistent; there the FATT is recovered at the level of the scalar functional through a Neyman-orthogonal estimating equation, again yielding a valid influence function (Section 5).

3 Forward-Looking Estimands

3.1 The Policy Evaluation Challenge

Policy evaluation is fundamentally future-oriented. A policymaker considering whether to maintain or repeal an existing policy needs to predict: What will be the effect going forward? A jurisdiction considering adoption needs: What would be the effect if we implement it now?

Standard estimands do not directly answer these questions. The overall ATT, for example, averages treatment effects over past periods $t \leq p$. If a policy was implemented over years 2010–2020, the ATT tells us the average effect during those years. But a policymaker in 2021 deciding whether to continue the policy needs to know: What will the effect be in 2021? When effects are dynamic, past averages may poorly predict future effects.

Consider three examples: A state evaluating whether to maintain expanded Medicaid coverage needs effects for the next budget cycle, not averages over the past decade. A city deciding whether to continue a policing reform needs effects under current conditions, not historical averages. A school district considering whether to expand a tutoring program needs effects for next year’s cohort, not past cohorts. In each case, the policy-relevant estimand targets the future, not the past.

We define a family of three future causal effects—the Future ATT, Future ATU, and Future ATE—that map directly onto the three decisions a policymaker faces: whether to *repeal* a policy already in place, whether to *adopt* one not yet in place, and the effect of *universal* adoption. As we show in Sections 4–5, the three share identification machinery, differing only in the population over which the effect is averaged.

3.2 Future Average Treatment Effect on the Treated (FATT)

We define the **Future ATT (FATT)** at horizon m as:

$$\tau_{\text{FATT}}(p + m) = \mathbb{E}_{P_{p+m}} \{Y_{i,p+m}(1) - Y_{i,p+m}(0) \mid A_{ip} = 1\}, \quad (4)$$

where P_{p+m} denotes the distribution at future time $p + m$, and the conditioning is on treatment status at period p (the end of the observed data).

Policy Relevance The FATT answers: “What will be the treatment effect at time $p + m$ for units that have already adopted the policy?” This is the estimand needed when deciding whether to maintain or repeal an existing policy.

Connection to Group-Time ATTs The FATT aggregates effects across adoption cohorts at the future time:

$$\tau_{\text{FATT}}(p+m) = \sum_g \pi_g \theta_{g,p+m}, \quad (5)$$

where $\pi_g = P(G_i = g \mid A_{ip} = 1)$ is the proportion of treated units from cohort g , and $\theta_{g,p+m}$ is the group-time ATT at the future period $p+m$. The challenge: $\theta_{g,p+m}$ is unobserved for $p+m > p$.

Contrast with Standard ATT The standard ATT averages effects over observed periods:

$$\theta_{\text{ATT}} = \mathbb{E}_{t \leq p} \{\theta_{gt} \mid A_{it} = 1\}. \quad (6)$$

If effects are constant over time, $\tau_{\text{FATT}}(p+m) = \theta_{\text{ATT}}$. But if effects are dynamic, the standard ATT—a past average—may poorly predict the future effect.

3.3 Future Average Treatment Effect on the Untreated (FATU)

We define the **Future ATU (FATU)** at horizon m as

$$\tau_{\text{FATU}}(p+m) = \mathbb{E}_{p_{p+m}} \{Y_{i,p+m}(1) - Y_{i,p+m}(0) \mid A_{ip} = 0\}, \quad (7)$$

the future effect among units *not* treated by the end of the observed window. The FATU answers the adoption question: “What would the effect be at time $p+m$ for units that have not yet adopted the policy, were they to adopt it?” It is the relevant estimand for a jurisdiction deciding whether to implement a policy currently in place elsewhere, and is the future-looking analog of the average treatment effect on the untreated (ATU).

3.4 Future Average Treatment Effect (FATE)

We define the **Future ATE (FATE)** at horizon m as:

$$\tau_{\text{FATE}}(p+m) = \mathbb{E}_{p_{p+m}} \{Y_{i,p+m}(1) - Y_{i,p+m}(0)\}. \quad (8)$$

Policy Relevance The FATE answers: “What would be the average treatment effect at time $p+m$ if the policy were implemented universally?” This estimand is relevant when considering expanding a policy to the full population.

Relationship to FATT and FATU The FATE is the population mixture of the other two:

$$\tau_{\text{FATE}}(p+m) = \lambda \tau_{\text{FATT}}(p+m) + (1-\lambda) \tau_{\text{FATU}}(p+m), \quad (9)$$

where $\lambda = P(A_{ip} = 1)$. When there is no selection on treatment effect heterogeneity, $\tau_{\text{FATE}} = \tau_{\text{FATT}}$ and all three coincide. In general they differ, and the gap between the FATT and FATU is exactly what distinguishes the repeal question (effects among current adopters) from the adoption question (effects among current non-adopters).

3.5 The Identification Challenge

To identify the FATT, FATU, and FATE, we must bridge from the data we can analyze (the past) to the estimands we care about (the future). All three involve effects at time $p+m > p$, beyond our observed data: we cannot directly estimate $\theta_{g,p+m}$ or the future potential outcomes $Y_{i,p+m}(a)$ because they correspond to periods we have not yet observed.

Identification requires assumptions that connect the future to the past. Section 4 presents three strategies that differ in what they assume is invariant across time: effects themselves (Path 1), temporal patterns (Path 2), or structural covariate relationships (Path 3). We develop each primarily for the FATT and indicate how the FATU and FATE follow by changing the population over which the effect is averaged.

4 Identification via Extrapolation Functions

4.1 General Framework

All three paths follow a common template: plug an existing, asymptotically linear estimator of some *relevant object* into a map to the future estimand, and propagate that estimator's influence function for inference. What differs is the relevant object and the invariance assumption that licenses the extrapolation. We introduce no new first-stage estimator; the contribution is the forward map and the uncertainty propagation.

For Paths 1 and 2 the relevant object is the group-time ATTs θ_{gt} , and we posit an **extrapolation function** $h(g, t; \gamma)$ with $h(g, t; \gamma) = \theta_{gt}$ for observed times $t \leq p$ and extending the pattern to future times $t > p$; γ is identified from $\{\theta_{gt} : t \leq p\}$ and the FATT is $\sum_g \pi_g h(g, p+m; \gamma)$. For Path 3 the relevant object is instead the covariate-conditional effect $\tau(x)$, estimated directly from the unit-level data and transported to the future covariate distribution; it does not pass through the marginal θ_{gt} at all. The three paths differ as follows:

- **Path 1:** $h(g, t; \gamma) = \theta_g$ (effects time-invariant within cohort).
- **Path 2:** $h(g, t; \gamma) = f(g, t; \gamma)$ for a known functional form f (temporal pattern extrapolated).

- **Path 3:** $\tau_{\text{FATT}}(p+m) = \int \tau(x) dF_{X|A_{ip}=1}^{p+m}(x)$ (conditional effect held fixed, covariate distribution transported).

4.2 Path 1: Time Homogeneity

Assumption 6 (Strict Time Homogeneity). *For all cohorts g and times t ,*

$$\theta_{gt} = \theta_g. \quad (10)$$

Under Assumption 6, treatment effects are constant across calendar time for each cohort. This is the most restrictive path but requires minimal modeling.

Proposition 4.1. *Under Assumption 6,*

$$\tau_{\text{FATT}}(p+m) = \sum_g \pi_g \theta_g. \quad (11)$$

Proof. See Appendix A.2. □

Interpretation Proposition 4.1 identifies the FATT as a weighted average of cohort-specific ATTs. This assumption is plausible when treatment effects are driven by stable characteristics (e.g., time since treatment) rather than calendar time. It holds when policy implementation does not change over time and there are no secular trends affecting treatment effects.

Testable Implications Assumption 6 has testable implications: we can compare θ_{gt} across observed times $t \leq p$ for each group g . If effects vary substantially with calendar time in the observed data, time homogeneity is unlikely to hold for future periods. Testing approaches analogous to parallel trends tests apply here [Dette and Schumann, 2024].

Weaker Aggregate Homogeneity

Assumption 7 (Aggregate Time Homogeneity). *The overall ATT is constant over time:*

$$\tau_{\text{FATT}}(p+1) = \theta_{\text{ATT}}. \quad (12)$$

Assumption 7 is weaker than strict time homogeneity: it allows group-time variation that averages out. Under this assumption, the backward-looking ATT directly identifies the forward-looking FATT. This is the implicit assumption when researchers estimate an ATT and discuss policy implications as if the effect is stable [Gabler et al., 2009].

4.3 Path 2: Parametric Temporal Models

Assumption 8 (Parametric Extrapolation). *For a known function $f(\cdot)$ and unknown parameter γ ,*

$$\theta_{gt} = f(g, t; \gamma) \text{ for all } g, t. \quad (13)$$

Under Assumption 8, effects follow a parametric model. Common specifications include:

- **Linear trend:** $f(g, t; \gamma) = \alpha_g + \beta(t - g)$
- **Quadratic:** $f(g, t; \gamma) = \alpha_g + \beta_1(t - g) + \beta_2(t - g)^2$
- **Spline:** $f(g, t; \gamma)$ with knots at specified event-times

Proposition 4.2. *Under Assumption 8 with γ identified from $\{\theta_{gt} : t \leq p\}$,*

$$\tau_{FATT}(p + m) = \sum_g \pi_g f(g, p + m; \hat{\gamma}). \quad (14)$$

Proof. See Appendix A.3. □

Estimation Estimation under Path 2 proceeds in four steps:

1. Obtain first-stage estimates $\hat{\theta}_{gt}$ for $t \leq p$
2. Fit the parametric model: estimate $\hat{\gamma}$ by regressing $\hat{\theta}_{gt}$ on $f(g, t; \gamma)$
3. Extrapolate: compute $f(g, p + m; \hat{\gamma})$ for future periods
4. Aggregate: $\hat{\tau}_{FATT}(p + m) = \sum_g \hat{\pi}_g f(g, p + m; \hat{\gamma})$

Model Selection The challenge: we don't know the correct functional form $f(\cdot)$. In-sample fit favors overparameterized models, which may extrapolate poorly. Section 5.2 develops a time-series cross-validation framework to select models based on extrapolation performance.

When Path 2 Applies Path 2 is appropriate when effects exhibit recognizable temporal patterns (e.g., effects grow over time, fade out, stabilize). The key assumption is that the functional form $f(\cdot)$ captures the true dynamics and extends beyond the observed window. Misspecification leads to biased extrapolation.

4.4 Path 3: Covariate Transport

Paths 1 and 2 both rest on *reduced-form* stability: effects are either constant or follow a smooth temporal trend. Both fail under *regime change*—structural shifts in the policy environment or economic conditions that break historical patterns. The Lucas Critique [Lucas Jr, 1976] warns precisely that reduced-form relationships estimated under one regime need not survive a change of regime. Path 3 instead holds fixed a *conditional* object—the effect for units with covariate value x —and lets the marginal effect move only through the covariate distribution. Crucially, and unlike Paths 1–2, Path 3 estimates this conditional effect directly from the unit-level data; it never inverts the marginal group-time ATTs.

Let $\tau_t(x) = \mathbb{E}_{\mathbb{P}_t} \{Y_{it}(1) - Y_{it}(0) \mid X_i = x, \text{treated}\}$ be the period- t covariate-conditional ATT, and write $\tau(x)$ for its common value under the stability assumption below.

Assumption 9 (First-stage conditional identification). *Under the researcher’s design, $\tau_t(x)$ is identified from the observed data ($t \leq p$) and estimable, with cross-fitting, at standard nonparametric rates—via a doubly-robust CATE learner under conditional unconfoundedness, or a doubly-robust conditional DiD estimator [Sant’Anna and Zhao, 2020] under conditional parallel trends.*

Assumption 10 (Structural covariate stability). *The conditional ATT is time-invariant: $\tau_t(x) = \tau(x)$ for all x and all $t \in \{1, \dots, p, p + m\}$.*

Assumption 11 (Target access and overlap). *The future treated covariate distribution $F_{X|A_{ip}=1}^{p+m}$ is accessible (observed under time-invariant covariates, or forecasted), and the density ratio $w(x) = dF_{X|A_{ip}=1}^{p+m} / dF_X^{\text{src}}(x)$ is bounded.*

Proposition 4.3. *Under Assumptions 9–11,*

$$\tau_{\text{FATT}}(p + m) = \int \tau(x) dF_{X|A_{ip}=1}^{p+m}(x) = \mathbb{E}_{\text{src}} \{w(X) \tau(X)\}. \quad (15)$$

Proof. The first equality is iterated expectations; Assumption 10 identifies τ from observed periods; the second is change of measure, valid since w exists. No injectivity condition is needed because τ is estimated directly, not recovered from the marginals. See Appendix A.4. $\square$

Estimation. Fit $\hat{\tau}(\cdot)$ on the unit-level data with any asymptotically linear / cross-fit learner (a doubly-robust parametric model, or a nonparametric learner such as a causal forest), then average over a target covariate sample: $\hat{\tau}_{\text{FATT}}(p + m) = n^{*-1} \sum_i \hat{\tau}(X_i^*)$. Because $\hat{\tau}$ need not be $\sqrt{n}$ -consistent, inference is handled at the level of the scalar functional via a doubly-robust score (Section 5).

Lucas Critique connection. Path 3 is the Lucas Critique’s prescription made operational: hold fixed the structural object ($\tau(x)$, the “deep parameter”) and let the reduced-form marginal effect change only through the covariate distribution. The

effect of a policy may depend on local economic or demographic conditions (covariates) rather than calendar time per se; even when aggregate patterns shift, $\tau(x)$ may persist. This makes Path 3 more robust than Paths 1–2 to regime change—provided the shift is captured by observed covariates.

When Path 3 applies. Path 3 requires (i) covariates that capture the drivers of effect heterogeneity, (ii) credibility of conditional stability, and (iii) overlap (bounded w): the future covariate support must be covered by the observed data. When heterogeneity is driven by unobserved factors, or the future moves into covariate regions never observed, Path 3 fails. Assumption 10 is testable across observed periods (compare $\hat{\tau}_t(\cdot)$ for $t \leq p$) though untestable at $p + m$.

FATU and FATE. Replacing the target distribution $F_{X|A_{ip}=1}^{p+m}$ by the untreated units’ or the full population’s covariate distribution identifies the FATU and FATE, provided $\tau(x)$ also describes those units—a cross-group transportability (no-effect-modification-beyond-X) assumption that the FATT itself does not require.

4.5 Synthesis: Comparing the Three Paths

Table 1 compares the three identification paths. Each plugs an existing estimator into a forward map under a different invariance assumption; they are parallel strategies, not competitors.

Table 1: Three Identification Paths for the FATT

Path	First-stage object	Invariance	Advantage	Disadvantage
1 (Homogeneity)	θ_{gt}	Effects constant over time	Minimal modeling	Very restrictive
2 (Parametric)	θ_{gt}	Temporal form known	Explicit dynamics	Misspecification
3 (Transport)	$\tau(x)$	Conditional effect stable	Robust to regime change	Needs covariates + overlap

Guidance for Practice

- Use **Path 1** if time homogeneity is plausible (e.g., stable policy implementation, short horizons).
- Use **Path 2** when temporal patterns are evident in observed data and expected to continue.
- Use **Path 3** when regime change is anticipated or covariates capture structural drivers.

The paths are not mutually exclusive. Researchers can report results under multiple paths as sensitivity analysis, or combine paths (e.g., covariate-specific temporal trends).

5 Semiparametric Inference

5.1 Efficient Influence Function Propagation

The FATT is a functional of the first-stage estimates $\{\theta_{gt}\}$ and, for Paths 2–3, additional parameters (γ or $\theta(x, g)$). Standard errors must account for sampling variability in all components. We use the efficient influence function (EIF) framework to propagate uncertainty.

Theorem 5.1. *Under regularity conditions (Appendix A.1),*

$$\sqrt{n}(\hat{\tau}_{FATT} - \tau_{FATT}) \xrightarrow{d} N(0, V), \quad (16)$$

where $V = \mathbb{E}[\psi_{FATT}^2]$ and ψ_{FATT} is the efficient influence function (Appendix B).

Proof. See Appendix A.5. □

Implementation The R package `extrapolateATT` computes EIFs and standard errors for all three paths. See Appendix C.4 for code examples.

5.2 Model Selection for Path 2

When using Path 2, we face a model selection problem: which functional form $f(\cdot)$ is correct? In-sample fit (e.g., R^2 on $t \leq p$) favors complex models that may extrapolate poorly. We develop a time-series cross-validation (CV) framework that evaluates models based on out-of-sample extrapolation.

Time-Series Cross-Validation

1. Split observed periods: training ($t \leq p - h$) and validation ($p - h < t \leq p$).
2. For each candidate model $f_k(\cdot)$, fit $\hat{\gamma}_k$ on training data.
3. Predict validation-period effects: $\tilde{\theta}_{gt,k} = f_k(g, t; \hat{\gamma}_k)$ for $p - h < t \leq p$.
4. Compute validation metric (e.g., mean squared error, coverage).
5. Select the model with best out-of-sample performance.

Coverage-Based Selection We recommend a coverage-based criterion: select the model whose 95% confidence intervals achieve approximately 95% coverage on the validation set. This penalizes overconfident extrapolation and balances bias and uncertainty. Details in Appendix C.

Post-Selection Inference Standard inference treats the selected model as fixed, leading to undercoverage. We discuss adjustments (sample splitting, post-selection bootstrap) in Appendix C.3.

6 Simulations

6.1 Design

We conduct simulations with $n = 500$ units, $p = 20$ periods, and 5 adoption cohorts ($G \in \{5, 8, 11, 14, 17\}$). We estimate the FATT at $p + 1 = 21$ using all three paths. First-stage estimates are obtained via difference-in-differences with parallel trends (known to hold in the DGP). We report bias, root mean squared error (RMSE), and coverage of 95% confidence intervals over 1000 replications. Full DGP details are in Appendix D.

6.2 Three Core Scenarios

- **Scenario 1 (Homogeneity):** $\theta_{gt} = \alpha_g$ (constant over t). Path 1 is correctly specified.
- **Scenario 2 (Linear Dynamics):** $\theta_{gt} = \alpha_g + \beta(t - g)$. Path 2 with linear model is correct.
- **Scenario 3 (Regime Change):** $\theta(x, g)$ stable but aggregate θ_{gt} varies due to shifting $P_{X|t}$. Path 3 is correct.

6.3 Results

Table 2 presents the main simulation results.

Table 2: Simulation Results: Bias, RMSE, and Coverage				
Scenario	Method	Bias	RMSE	Coverage
1: Homogeneity	Path 1	0.01	0.12	0.95
1: Homogeneity	Path 2	0.02	0.14	0.94
1: Homogeneity	Path 3	0.03	0.16	0.95
2: Linear Dynamics	Path 1	-0.45	0.48	0.76
2: Linear Dynamics	Path 2 (CV)	0.03	0.19	0.94
2: Linear Dynamics	Path 2 (wrong)	-0.22	0.27	0.87
3: Regime Change	Path 1	-0.61	0.64	0.68
3: Regime Change	Path 2	-0.38	0.43	0.81
3: Regime Change	Path 3	0.05	0.21	0.93

Interpretation

- **Scenario 1:** All three paths perform well when homogeneity holds. Path 1 is most efficient (lowest RMSE).

- **Scenario 2:** Path 1 fails dramatically (bias = -0.45, coverage = 0.76) when dynamics are present. Path 2 with cross-validation succeeds. Model selection matters: using the wrong parametric form (e.g., quadratic when truth is linear) degrades performance but is less severe than ignoring dynamics entirely.
- **Scenario 3:** Only Path 3 succeeds under regime change. Reduced-form extrapolation (Paths 1–2) fails because aggregate patterns break. Structural covariate stability enables identification.

Key Takeaway No single path dominates across all scenarios. The appropriate choice depends on which invariance assumption holds in the setting. When uncertain, researchers should report results under multiple paths or use diagnostic checks (e.g., testing time homogeneity on observed data).

Extended simulation results, including small-sample properties, longer horizons, and stress tests, are in Appendix D.

7 Application: Stand-Your-Ground Laws

7.1 Background

Stand-Your-Ground (SYG) laws remove the duty to retreat before using deadly force in self-defense. Between 2005 and 2015, over 25 US states adopted such laws. The effects on homicide rates are controversial [?]. We use state-level data from 1981–2022 to illustrate forward-looking causal inference. We treat 1981–2015 as the training period and 2016–2022 as a validation period (holding out future data to assess extrapolation performance).

7.2 Design and Estimands

We estimate the FATT at three future time points: 2016, 2020, and 2022. The question: What is the effect of SYG laws on homicide rates in these years for states that adopted the laws?

First-Stage Estimation We use the [Callaway and Sant’Anna \[2021\]](#) difference-in-differences estimator to obtain group-time ATTs for the training period (1981–2015), assuming parallel trends.

Three Paths We apply all three paths:

1. **Path 1:** Assume time homogeneity; aggregate group-specific effects.
2. **Path 2:** Fit a linear event-time trend to $\{\hat{\theta}_{gt}\}$ and extrapolate to 2016–2022.
3. **Path 3:** Model conditional effects as a function of baseline state covariates (poverty rate, urbanization) and integrate over 2016 covariate distributions.

7.3 Validation Results

Table 3 compares predicted FATT (from each path, using only 1981–2015 data) to the “observed” effect (estimated using 2016–2022 data).

Year	Observed Effect	Path 1	Path 2	Path 3
2016	0.45 (0.12)	0.22 (0.10)	0.28 (0.13)	0.31 (0.14)
2017	0.52 (0.13)	0.22 (0.10)	0.35 (0.14)	0.38 (0.15)
2018	0.58 (0.14)	0.22 (0.10)	0.42 (0.15)	0.44 (0.16)
2019	0.63 (0.14)	0.22 (0.10)	0.49 (0.16)	0.51 (0.17)
2020	0.71 (0.15)	0.22 (0.10)	0.56 (0.17)	0.58 (0.18)

Note: Standard errors in parentheses. Observed effects estimated on held-out data (2016–2022). All three paths use only training data (1981–2015) for prediction.

Findings All three methods underpredict the validation-period effects:

- **Path 1** assumes constant effects and predicts 0.22 for all years. Actual effects grow from 0.45 (2016) to 0.71 (2020).
- **Path 2** incorporates a linear trend and performs better, but still underpredicts by 0.15–0.20 in later years.
- **Path 3** with covariates performs best but still misses by 0.10–0.15.

7.4 Promise and Limits

This application illustrates both the promise and limits of temporal extrapolation.

Promise The methods succeeded in several ways. First, Path 3 (covariate integration) substantially outperformed reduced-form approaches, demonstrating that structural modeling can improve extrapolation when covariates capture drivers of heterogeneity. Second, all three paths provided reasonable uncertainty quantification—confidence intervals reflected estimation uncertainty, even if they failed to capture model misspecification. Third, Paths 2 and 3 partially captured temporal trends, getting closer to validation-period effects than naive time-homogeneity assumptions (Path 1).

Limits Where extrapolation failed: All methods underpredicted validation-period effects, with gaps widening over time. Even Path 3, which performed best, missed by 0.10–0.15 in later years. Possible explanations include: (1) effects accelerated post-2015 faster than parametric models captured; (2) regime change not fully reflected in baseline covariates (e.g., cultural shifts, spillover effects, 2020 pandemic disruptions); (3) general equilibrium or feedback effects not modeled. The failure was not merely imprecision—it was systematic underprediction, suggesting extrapolation assumptions were violated.

Lesson Honest reporting of both successes and failures strengthens the contribution. This application shows that even sophisticated extrapolation methods rest on untestable assumptions about the future. When validation data are available—as here, where we held out 2016–2022—they reveal when assumptions break. In real policy settings, such validation is rarely possible, underscoring the inherent uncertainty in forward-looking causal inference. Transparent reporting about limitations is not a weakness but a feature of credible science.

Additional application details (data sources, robustness checks) are in Appendix E.

8 Discussion

8.1 Summary

We formalized the gap between backward-looking estimands (standard ATT, group-time effects) and forward-looking policy questions through two new estimands: the Future ATT (FATT) and Future ATE (FATE). These estimands target treatment effects at future time periods and are the relevant quantities for policy evaluation.

We presented three identification paths that differ in what they assume is invariant across time. Path 1 assumes effects are constant, Path 2 models temporal dynamics parametrically, and Path 3 leverages structural stability of covariate-conditional effects. We developed semiparametric inference via efficient influence function propagation and a model selection framework using time-series cross-validation.

Simulations show that no path universally dominates: the best choice depends on which invariance assumption holds. Our application to Stand-Your-Ground laws illustrates the challenges of extrapolation in real settings, with all methods underpredicting validation-period effects.

8.2 Guidance for Practice

Choosing a Path

- Use **Path 1** when time homogeneity is plausible (e.g., stable implementation, short horizons, no evident trends in observed data).
- Use **Path 2** when temporal patterns are clear in observed data and expected to continue. Employ model selection via cross-validation.
- Use **Path 3** when effect heterogeneity is driven by observable covariates and regime change is anticipated (e.g., policy reform, economic shifts).

Diagnostic Checks Test time homogeneity on observed data: compare θ_{gt} across $t \leq p$. If effects vary substantially in the training period, extrapolation requires Path 2 or 3. Report results under multiple paths when uncertain (sensitivity analysis).

Transparent Reporting Be explicit about invariance assumptions. Report uncertainty. Do not oversell extrapolation as prediction—it rests on untestable assumptions about the future.

8.3 Limitations

Our framework has several limitations:

- **Relies on structural stability:** All three paths assume some aspect of the causal structure extends from the training period to future periods. When this fails (as in our application), extrapolation is biased.
- **Single future period:** We focus on FATT at a single horizon $p + m$. Extending to multiple periods or forecasting entire trajectories is left for future work.
- **No formal sensitivity analysis:** We do not provide sensitivity bounds for violations of extrapolation assumptions. Developing such bounds is an important direction.
- **First-stage identification:** We assume θ_{gt} are identified. If first-stage assumptions (e.g., parallel trends) fail, bias propagates to the FATT.

8.4 Extensions

Future research directions include:

- **Multiple future periods:** Forecasting $\tau_{\text{FATT}}(p + 1), \tau_{\text{FATT}}(p + 2), \dots$ jointly.
- **Sensitivity analysis:** Bounding the FATT under partial violations of extrapolation assumptions.
- **Dynamic treatment regimes:** Extending to time-varying policies where treatment can be turned on/off.
- **Subgroup effects:** Estimating covariate-specific FATTs for targeted policy recommendations.

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